

RegRadar

A production-grade, multi-agent regulatory intelligence system that monitors SEC, FDA, and FINRA filings in real time — triaging risk, generating plain-English briefs, and alerting the right people instantly.

Multi-Agent RAG System | AI Engineering Portfolio | Solo / Small Team | 8–12 Weeks

1. PRODUCT OVERVIEW

RegRadar is an AI-powered regulatory intelligence platform designed for compliance teams at mid-market companies in finance, healthcare, and energy. It continuously ingests regulatory filings from SEC EDGAR, FDA, and FINRA, routes them through a multi-agent pipeline to triage priority, extract risk signals, generate plain-English impact briefs, and deliver targeted alerts to the right stakeholder — all within minutes of a filing's publication. The system exposes a production REST API, maintains a RAG-enabled knowledge base of all processed filings, and ships with a full LLM evaluation harness.

2. PROBLEM STATEMENT

Compliance teams at mid-market companies spend 20–40 hours per week manually reading regulatory filings. Missing a single material SEC rule change or FDA guidance update can trigger costly legal reviews, product recalls, or board-level escalations — yet most organizations only learn about new filings days later via Google Alerts or industry newsletters. There is no automated system that reads, understands, prioritizes, and routes regulatory documents in real time.

- No real-time monitoring: filings are discovered manually, hours or days late
- No intelligent triage: all filings are treated equally regardless of business impact
- No structured extraction: key obligations, deadlines, and risk flags buried in 80-page PDFs
- No institutional memory: past rulings are unsearchable, forcing repeated legal research
- No targeted routing: the wrong person gets alerted, or nobody does

3. GOALS & NON-GOALS

Goals

G1	Ingest new regulatory filings within 5 minutes of publication
G2	Classify filing domain (financial, clinical, environmental) with >90% accuracy using zero-shot
G3	Generate a 3–5 sentence plain-English impact brief per filing
G4	Assign a risk score (Low / Medium / High / Critical) based on configurable rules
G5	Deliver alerts to the correct stakeholder via Slack and email within 3 minutes of ingestion
G6	Enable natural language Q&A; over the full corpus of processed filings via RAG
G7	Expose all functionality via a versioned REST API with <500ms P99 latency
G8	Ship with an automated eval harness tracking Ragas faithfulness, ROUGE-L, and alert precision

Non-Goals

NG1	Not a legal advisory tool — output is intelligence, not legal counsel
NG2	Not a document management system — raw PDFs are stored externally (S3)
NG3	No UI dashboard in V1 — API-first; a simple React dashboard is a V2 stretch goal
NG4	Does not cover international regulators (EU, FCA) in V1

4. USER PERSONAS

Persona	Needs	Primary Touch Point
Chief Compliance Officer	Needs high-priority alerts on filings that require board-level action. Does not read full documents. Wants a 3-sentence brief and a risk score.	Critical filings only (High/Critical)
Compliance Analyst	Reads all medium and high filings in detail. Needs structured extraction: obligations, deadlines, affected products. Uses the API to build internal reports.	All filings with full extraction
Legal Counsel	Needs to search past rulings to inform current advice. Uses RAG Q&A; to ask 'What has SEC said about X over the past 3 years?'	RAG Q&A; over full corpus
Engineering Lead	Integrates RegRadar into internal tooling via the REST API. Needs reliable webhooks, versioned endpoints, and clear error contracts.	API consumer — webhooks + REST

5. USER STORIES

ID	Persona	I want to...	So that...	Pri
US-01	CCO	receive a Slack alert within 3 min of a Critical filing	I can act before my peers hear about it	P0
US-02	Analyst	see extracted obligations and deadlines from a filing	I don't have to read 80 pages to find the key requirements	P0
US-03	Analyst	get a risk score (Low/Med/High/Critical) per filing	I can prioritize my review queue	P0
US-04	Legal	ask 'What has SEC ruled on crypto disclosures since 2023?'	I can research precedents without manual searching	P0
US-05	Eng Lead	receive a webhook POST when a new filing is processed	I can trigger downstream workflows automatically	P0
US-06	Analyst	filter filings by domain, date range, and risk level via API	I can build custom compliance dashboards	P1
US-07	CCO	get a weekly digest of all Critical and High filings	I have a running record for board reporting	P1
US-08	Analyst	compare a new filing against similar past filings	I can spot what changed vs what we already complied with	P1
US-09	Eng Lead	see cost-per-filing tracked in the observability dashboard	I can forecast API spend at scale	P2
US-10	Any	configure which regulators and domains to monitor	I only get alerts relevant to my industry	P2

6. SYSTEM ARCHITECTURE — MULTI-AGENT PIPELINE

RegRadar is built on a LangGraph-orchestrated multi-agent DAG. A supervisor agent coordinates 5 specialized sub-agents that share a typed state object. Each agent has a single responsibility, can be individually tested, and is independently swappable (e.g. replacing the summarization model without touching retrieval).

Agent	Powered By	Responsibility
Ingestion Agent	Prefect scheduler + SEC EDGAR API, FDA RSS, FINRA feeds	Polls data sources every 5 min, downloads new filings, stores raw PDF to S3, enqueues job in Redis
Triage Agent	HF Zero-Shot Classification (facebook/bart-large-mnli)	Classifies filing domain, assigns preliminary priority score, routes to correct analyst persona queue
RAG Retrieval Agent	LlamaIndex + pgvector + HF Feature Extraction embeddings	Chunks the filing PDF, embeds with text-embedding-3-small, retrieves similar past filings for context
Analysis Agent	GPT-4o with structured output (JSON mode)	Extracts obligations, deadlines, risk flags, affected products, competitor mentions into typed schema
Summarization Agent	HF Summarization model (facebook/bart-large-cnn) + LLM	Generates 3-sentence plain-English brief, risk score label, and per-persona tailored summaries
Delivery Agent	FastAPI webhooks + Slack API + SendGrid	Routes briefs to correct stakeholder, fires webhook POSTs, queues weekly digest, logs all deliveries

7. DATA FLOW

Step 1

Ingestion Agent polls SEC EDGAR, FDA, FINRA every 5 minutes. New filings detected via ETag / last-modified headers.

Step 2

Raw PDF stored to S3. Metadata (source, timestamp, filing type, entity name) written to Postgres filings table.

Step 3

Job pushed to Redis queue. Triage Agent picks it up, runs zero-shot classification, writes domain + priority to DB.

Step 4

RAG Retrieval Agent chunks the PDF (512-token chunks, 50-token overlap), embeds each chunk, stores vectors in pgvector. Retrieves top-5 similar past filings for context window.

Step 5

Analysis Agent calls GPT-4o with structured output schema. Extracts: obligations (list), deadlines (list of dates), risk_level (enum), affected_products (list), key_entities (list).

Step 6

Summarization Agent generates: executive_brief (3 sentences), cco_summary, analyst_summary, engineer_summary.

Step 7

Delivery Agent: fires Slack message to configured channel, sends webhook POST to all registered endpoints, queues email via SendGrid, logs delivery receipt.

8. TECH STACK

Layer	Technology	Justification
Orchestration	LangGraph 0.2	Typed state graph, conditional edges, human-in-the-loop support
LLM — Analysis	GPT-4o (structured output / JSON mode)	Best-in-class for complex extraction with typed schema enforcement
LLM — Summarization	HF bart-large-cnn + Granite fallback	Cost-efficient for summarization; Granite as cheap fallback on low-priority
Classification	HF facebook/bart-large-mnli	Zero-shot classification — no labeled training data required
Embeddings	text-embedding-3-small (OpenAI)	1536-dim, fast, cheap; used for RAG retrieval chunks
Vector Store	pgvector (Postgres extension)	Single DB for relational + vector; avoids managing a separate Pinecone instance
RAG Framework	LlamaIndex 0.10	Best chunking primitives; built-in re-ranking and hybrid search
Ingestion Scheduler	Prefect 2.x	Cron + retry logic + observability for data pipeline flows
Job Queue	Redis + Celery	Async processing; decouples ingestion from agent pipeline

API	FastAPI + Pydantic v2	Typed request/response models; async; automatic OpenAPI docs
Storage	Supabase (Postgres) + S3	Managed Postgres + pgvector; S3 for raw PDF storage
Eval	Ragas + ROUGE + custom F1	Faithfulness, context recall, ROUGE-L on summaries, precision on alerts
Observability	LangSmith + Langfuse	Prompt versioning, cost-per-filing, latency tracking, A/B model compare
Delivery	Slack Webhooks + SendGrid + FastAPI	Multi-channel alert routing with delivery receipts
Deployment	Docker + Fly.io	Containerized; Fly.io for cheap always-on worker + API hosting

9. API CONTRACT (KEY ENDPOINTS)

Method	Endpoint	Description
GET	/v1/filings	List all processed filings with filters: ?domain=financial&risk;=HIGH&since;=2026-01-01
GET	/v1/filings/{id}	Full filing record: brief, extracted obligations, deadlines, risk score, similar past filings
POST	/v1/filings/search	RAG-powered natural language search over all filings. Body: {query: string, top_k: int}
GET	/v1/filings/{id}/brief	Returns the persona-specific brief: ?persona=cco analyst engineer
POST	/v1/webhooks	Register a webhook URL to receive POST on every new processed filing
DELETE	/v1/webhooks/{id}	Deregister a webhook
GET	/v1/metrics	Eval metrics: Ragas faithfulness, alert precision/recall, avg latency, cost-per-filing
POST	/v1/config/sources	Configure which regulators/domains to monitor. Body: {sources: [SEC, FDA, FINRA], domains: [...]}

10. LLM EVALUATION HARNESS

Every prompt change triggers a regression suite before deployment. Metrics are logged to LangSmith and graphed over time. A failing eval blocks the deploy.

Metric	Tool	Target	What It Measures
Ragas Faithfulness	Ragas library	>0.87	Are brief claims grounded in the source filing?
Ragas Context Recall	Ragas library	>0.80	Did RAG retrieve the relevant past filings?
ROUGE-L	HF evaluate	>0.45	Summarization quality vs human-written reference briefs
Alert Precision	Custom test set (n=200)	<5% FP	How often does a 'Critical' alert turn out to be Low risk?
Alert Recall	Custom test set (n=200)	<3% FN	How often is a genuinely Critical filing missed or mislabeled?
Hallucination Rate	LangSmith + manual audit	<2%	Facts in brief that cannot be traced to source document
Extraction F1	Labeled obligation set	>0.82 F1	Precision/recall on extracted obligations and deadlines
P99 Latency	FastAPI middleware	<500ms	API response time at 99th percentile under normal load
Cost per Filing	LangSmith token tracking	<\$0.03	End-to-end LLM cost to process one filing

11. 12-WEEK BUILD PLAN

Weeks 1–2 — Foundation

Ingestion pipeline: SEC EDGAR + FDA RSS polling via Prefect. Raw PDF → S3. Postgres schema (filings, chunks, deliveries, webhook_registrations). Redis queue setup. Basic FastAPI skeleton with health check endpoint.

Weeks 3–4 — Triage + Classification Agent

Zero-shot classification agent with HF bart-large-mnli. Domain classification (financial / clinical / environmental / other). Priority scoring logic (keyword rules + LLM rerank). Write test set of 100 manually labeled filings for eval baseline.

Weeks 5–6 — RAG Layer

PDF chunking strategy: semantic chunking with 512-token max, 50-token overlap, section-boundary awareness. Embed all historical filings into pgvector. LlamaIndex retriever with hybrid BM25 + dense search. Measure retrieval quality: context recall on labeled test set.

Weeks 7–8 — Analysis + Summarization Agents

GPT-4o structured output schema: obligations (list), deadlines (date list), risk_level (enum), affected_products (list). Summarization agent with persona-specific brief templates. Prompt versioning in LangSmith. ROUGE-L regression suite.

Weeks 9–10 — Delivery + API

Delivery agent: Slack webhook, SendGrid email, webhook POST with HMAC signature. Full REST API: all endpoints, Pydantic v2 request/response models, OpenAPI docs. Webhook registration/deregistration. Weekly digest job.

Weeks 11–12 — LLMops, Eval Harness, Polish

Full Ragas eval suite integrated into CI. Cost-per-filing dashboard in Langfuse. A/B model routing (Granite vs GPT-4o for summarization based on priority tier). Architecture Decision Records (ADRs) written. Public README with system diagram. Load test to 100 concurrent filings. Deploy to Fly.io.

12. ARCHITECTURE DECISION RECORDS (ADRS)

ADR-01: pgvector over Pinecone

Decision: Use pgvector (Postgres extension) instead of a managed vector DB. Rationale: Regulatory data is already relational (filing metadata, obligations, deliveries). Keeping vector and relational data in one DB eliminates join complexity and reduces ops overhead. Tradeoff: pgvector is ~2x slower than Pinecone at 10M+ vectors, acceptable for this corpus size.

ADR-02: Zero-Shot over Fine-Tuned Classifier

Decision: Use HF zero-shot classification for domain triage instead of a fine-tuned model. Rationale: No labeled training data available at project start. Zero-shot achieves >90% accuracy on 4-class domain classification. Tradeoff: Lower accuracy than fine-tuned; plan to collect labels via human feedback loop and fine-tune in V2.

ADR-03: LangGraph over vanilla LangChain

Decision: Use LangGraph for multi-agent orchestration. Rationale: Typed state graph enables conditional routing (e.g. skip RAG for low-priority filings), retry logic per node, and human-in-the-loop interruption without custom state management. Tradeoff: Steeper learning curve; more boilerplate than a simple chain.

ADR-04: Tiered Model Routing

Decision: Route Critical/High filings to GPT-4o and Low/Medium to Granite-13B (HF). Rationale: GPT-4o is 10x more expensive than Granite. ~60% of filings are Low/Medium. Tiered routing cuts average cost-per-filing from \$0.12 to \$0.03 with <5% quality degradation on low-priority filings.

13. SUCCESS METRICS

<3 min	6 agents	>0.87	<\$0.03
Filing to Slack alert	LangGraph DAG	Ragas faithfulness	Cost per filing
500K+	>90%	<500ms	<2%
Filings indexed in V1	Classification accuracy	API P99 latency	Hallucination rate

14. RISKS & MITIGATIONS

Risk	Severity	Mitigation
R1 — LLM hallucination on legal text	HIGH	Source citations required in every extracted field. Ragas faithfulness >0.87 gates deploy. Human review queue for Critical filings.
R2 — SEC EDGAR rate limiting	MED	Respect 10 req/sec EDGAR limit. Exponential backoff in Prefect flow. Mirror to S3 immediately to avoid re-fetching.
R3 — PDF parsing quality	MED	Regulatory PDFs vary wildly (scanned, tabular, footnote-heavy). Maintain a test corpus of 50 edge-case PDFs. Fall back to OCR via pytesseract for scanned docs.
R4 — GPT-4o cost overrun	MED	Tiered routing (ADR-04) caps spend. Hard budget alert in Langfuse at \$50/day. Circuit breaker falls back to Granite if OpenAI is unavailable.
R5 — Zero-shot misclassification cascade	LOW	Misclassifying a Critical filing as Low delays the alert. Dual-model voting (bart-mnli + GPT-4o spot check) for all filings above a confidence threshold.

15. MVP SCOPE — SHIP VS CUT

SHIP (V1 MVP)	CUT (V2+)
✓ Ingestion: SEC EDGAR + FDA RSS (FINRA in V1.1)	✗ React dashboard UI
✓ Zero-shot domain classification + priority scoring	✗ Fine-tuned classification model

✓ RAG layer: pgvector + hybrid search	✗ International regulators (EU ESMA, FCA)
✓ Analysis extraction: obligations, deadlines, risk score	✗ PDF OCR for scanned legacy documents
✓ 3-sentence executive brief + persona summaries	✗ SSO / enterprise auth
✓ Slack + email delivery	✗ Bulk historical backfill >5 years
✓ Webhook registration + HMAC-signed POST	✗ iOS / email digest mobile app
✓ REST API (all P0 endpoints)	✗ Competitor mention trend analytics
✓ Ragas + ROUGE-L eval harness in CI	✗ Multi-tenant SaaS billing
✓ LangSmith prompt versioning + cost tracking	
✓ Deployment: Docker + Fly.io	

16. PORTFOLIO & RESUME VALUE

RegRadar demonstrates the full AI engineering stack that senior AI engineering roles require in 2026: multi-agent orchestration, production RAG, LLM evaluation, LLMops, and a real-world data integration — not just an LLM wrapper.

- ✓ Built a 6-agent regulatory intelligence system using LangGraph with conditional routing and typed shared state
- ✓ Designed and shipped a production RAG pipeline (LlamaIndex + pgvector) with hybrid BM25 + dense retrieval, achieving Ragas context recall >0.80
- ✓ Implemented tiered LLM routing (GPT-4o / Granite-13B) reducing cost-per-filing by 75% with <5% quality degradation
- ✓ Built an automated eval harness (Ragas, ROUGE-L, alert precision/recall) gating all prompt changes in CI
- ✓ Ingested and processed 500K+ real regulatory documents from SEC EDGAR and FDA with <3 min end-to-end latency
- ✓ Tracked prompt versioning, model A/B tests, and cost-per-filing in LangSmith + Langfuse
- ✓ Deployed production-grade FastAPI with async job queue (Celery + Redis), webhook delivery, and HMAC-signed callbacks